R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define the term Tokens in lexical analysis phase.	[2]
b)	Write a short note on Context Free Grammar.	[3]
c)	List different rules to compute FIRSTSET.	[2]
d)	Brief different types of LR parsers.	[3]
e)	List different data structures used for symbol table.	[2]
f)	Define syntax tree? Draw the syntax tree for the assignment statement. a :=b * -c + b * -c.	[3]
g)	Define DAG.	[2]
h)	Write a short note on global optimization.	[3]
i)	Write in detail the issues in the design of code generator.	[2]
j)	List the properties of code generation.	[3]

PART - B

(50 Marks)

2.a) Differentiate between the Pass and Phase in compiler construction.
b) Explain the general format of a LEX program with example. [5+5]

OR

3. Prepare the predictive parser for the following grammar:
S→a|b|(T)
T →T, S|S
Write down the necessary algorithms and define FIRST and FOLLOW. Show the behavior of the parser in the sentences.
a) (a,(a,a))
b) ((a,a),a,(a),a). [10]

4. State shift-reduce parsing? Explain in detail the conflicts that may occur during shift-reduce parsing. Illustrate with an example. [10]

OR

5. Construct SLR parsing table for the following grammar:
E→E+T/T
T→T*F/F
F→(E)/id. [10]

6.a) Explain Intermediate code generation for Basic block.
b) Write the advantages and disadvantages of heap storage allocation strategies. [5+5]

OR

7. Explain three address codes and mention its types. Generate the three-address code for the following C program fragment [10]

```
while(a > b)
{
if (c < d)
x = y + z;
else
x = y - z;
}
```

8.a) Explain the principle sources of code optimization in detail.
b) Write a brief note on Data Flow Analysis of structural programs. [5+5]

OR

9.a) Construct the DAG for the following basic block.

D:=B*C
E:=A+B
B:=B+C
A:=E-D

b) Explain in detail about Copy Propagation. [5+5]

10.a) Discuss various object code forms.
b) Write a short note on DAG for register allocation. [5+5]

OR

11. Explain Machine dependent code optimization in detail with an example. [10]

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